

```
In [2]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [4]: income_df = pd.read_csv(r'C:\Users\user\FSDS_Adv_Gene_AI_Agentic_AI\Statistics\Inc_Exp_Data.csv')
income_df
```

Out[4]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
0	5000	8000	3	2000	64200
1	6000	7000	2	3000	79920
2	10000	4500	2	0	112800
3	10000	2000	1	0	97200
4	12500	12000	2	3000	147000
5	14000	8000	2	0	196560
6	15000	16000	3	35000	167400
7	18000	20000	5	8000	216000
8	19000	9000	2	0	218880
9	20000	9000	4	0	220800
10	20000	18000	4	8000	278400
11	22000	25000	6	12000	279840
12	23400	5000	3	0	292032
13	24000	10500	6	0	316800
14	24000	10000	4	0	244800
15	25000	12300	3	0	246000
16	25000	20000	3	3500	261000
17	25000	10000	6	0	258000
18	29000	6600	2	2000	348000
19	30000	13000	4	0	385200
20	30500	25000	5	5000	351360
21	32000	15000	4	0	445440
22	34000	19000	6	0	330480
23	34000	25000	3	4000	469200
24	35000	12000	3	0	466200
25	35000	25000	4	0	449400
26	39000	8000	4	0	556920
27	40000	10000	4	0	412800
28	42000	15000	4	0	488880
29	43000	12000	4	0	619200
30	45000	25000	6	0	523800
31	45000	40000	6	3500	507600
32	45000	10000	2	1000	437400
33	45000	22000	4	2500	610200
34	46000	25000	5	3500	596160
35	47000	15000	7	0	456840

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
36	50000	20000	4	0	570000
37	50500	20000	3	0	581760
38	55000	45000	6	12000	600600
39	60000	10000	3	0	590400
40	60000	50000	6	10000	590400
41	65000	20000	4	5000	647400
42	70000	9000	2	0	756000
43	80000	20000	4	0	1075200
44	85000	25000	5	0	1142400
45	90000	48000	7	0	885600
46	98000	25000	5	0	1152480
47	100000	30000	6	0	1404000
48	100000	50000	4	20000	1032000
49	100000	40000	6	10000	1320000

In [5]:

income_df.head()

Out[5]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
0	5000	8000	3	2000	64200
1	6000	7000	2	3000	79920
2	10000	4500	2	0	112800
3	10000	2000	1	0	97200
4	12500	12000	2	3000	147000

In [6]:

income_df.head(2)

Out[6]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
0	5000	8000	3	2000	64200
1	6000	7000	2	3000	79920

In [7]:

income_df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                       50 non-null     int64
1   Mthly_HH_Expense                      50 non-null     int64
2   No_of_Fly_Members                     50 non-null     int64
3   Emi_or_Rent_Amt                       50 non-null     int64
4   Annual_HH_Income                      50 non-null     int64
5   Highest_Qualified_Member              50 non-null     object
6   No_of_Earning_Members                 50 non-null     int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB

```

In [8]:


```
income_df.shape
```

Out[8]: (50, 7)

In [9]:


```
income_df.columns
```

Out[9]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
'No_of_Earning_Members'],
dtype='object')

In [11]:


```
len(income_df.columns)
```

Out[11]: 7

In [10]:


```
income_df.describe()
```

Out[10]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Incom
count	50.000000	50.000000	50.000000	50.000000	5.000000e+C
mean	41558.000000	18818.000000	4.060000	3060.000000	4.900190e+C
std	26097.908979	12090.216824	1.517382	6241.434948	3.201358e+C
min	5000.000000	2000.000000	1.000000	0.000000	6.420000e+C
25%	23550.000000	10000.000000	3.000000	0.000000	2.587500e+C
50%	35000.000000	15500.000000	4.000000	0.000000	4.474200e+C
75%	50375.000000	25000.000000	5.000000	3500.000000	5.947200e+C
max	100000.000000	50000.000000	7.000000	35000.000000	1.404000e+C

In [13]:


```
income_df.describe().T      # T Referes to transpose
```

Out[13]:

	count	mean	std	min	25%	50%	75%	
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0	35000.0	50375.0	100000.0
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0	15500.0	25000.0	50000.0
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0	4.0	5.0	
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0	0.0	3500.0	35000.0
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0	447420.0	594720.0	1404000.0
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0	1.0	2.0	

In [14]:

income_df.isna()

Out[14]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
5	False	False	False	False	False
6	False	False	False	False	False
7	False	False	False	False	False
8	False	False	False	False	False
9	False	False	False	False	False
10	False	False	False	False	False
11	False	False	False	False	False
12	False	False	False	False	False
13	False	False	False	False	False
14	False	False	False	False	False
15	False	False	False	False	False
16	False	False	False	False	False
17	False	False	False	False	False
18	False	False	False	False	False
19	False	False	False	False	False
20	False	False	False	False	False
21	False	False	False	False	False
22	False	False	False	False	False
23	False	False	False	False	False
24	False	False	False	False	False
25	False	False	False	False	False
26	False	False	False	False	False
27	False	False	False	False	False
28	False	False	False	False	False
29	False	False	False	False	False
30	False	False	False	False	False
31	False	False	False	False	False
32	False	False	False	False	False
33	False	False	False	False	False
34	False	False	False	False	False
35	False	False	False	False	False

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income
36	False	False	False	False	False
37	False	False	False	False	False
38	False	False	False	False	False
39	False	False	False	False	False
40	False	False	False	False	False
41	False	False	False	False	False
42	False	False	False	False	False
43	False	False	False	False	False
44	False	False	False	False	False
45	False	False	False	False	False
46	False	False	False	False	False
47	False	False	False	False	False
48	False	False	False	False	False
49	False	False	False	False	False

```
In [15]: income_df.isna().any() # checks for null values
```

```
Out[15]: Mthly_HH_Income      False
Mthly_HH_Expense      False
No_of_Fly_Members     False
Emi_or_Rent_Amt       False
Annual_HH_Income      False
Highest_Qualified_Member False
No_of_Earning_Members False
dtype: bool
```

```
In [16]: income_df.columns
```

```
Out[16]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
               'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
               'No_of_Earning_Members'],
              dtype='object')
```

```
In [17]: print(income_df['Mthly_HH_Expense'].mean())
```

```
18818.0
```

```
In [18]: print(income_df['Mthly_HH_Expense'].median())
```

```
15500.0
```

```
In [21]: income_df['Mthly_HH_Expense'].mode()
```

```
Out[21]: 0    25000
Name: Mthly_HH_Expense, dtype: int64
```

```
In [25]: mth_exp_tmp = pd.crosstab(index = income_df['Mthly_HH_Expense'], columns = 'count')

mth_exp_tmp.reset_index(inplace = True)

mth_exp_tmp[mth_exp_tmp['count'] == income_df.Mthly_HH_Expense.value_counts().max()]
```

```
# Crosstab:
```

```
Out[25]: col_0  Mthly_HH_Expense  count
```

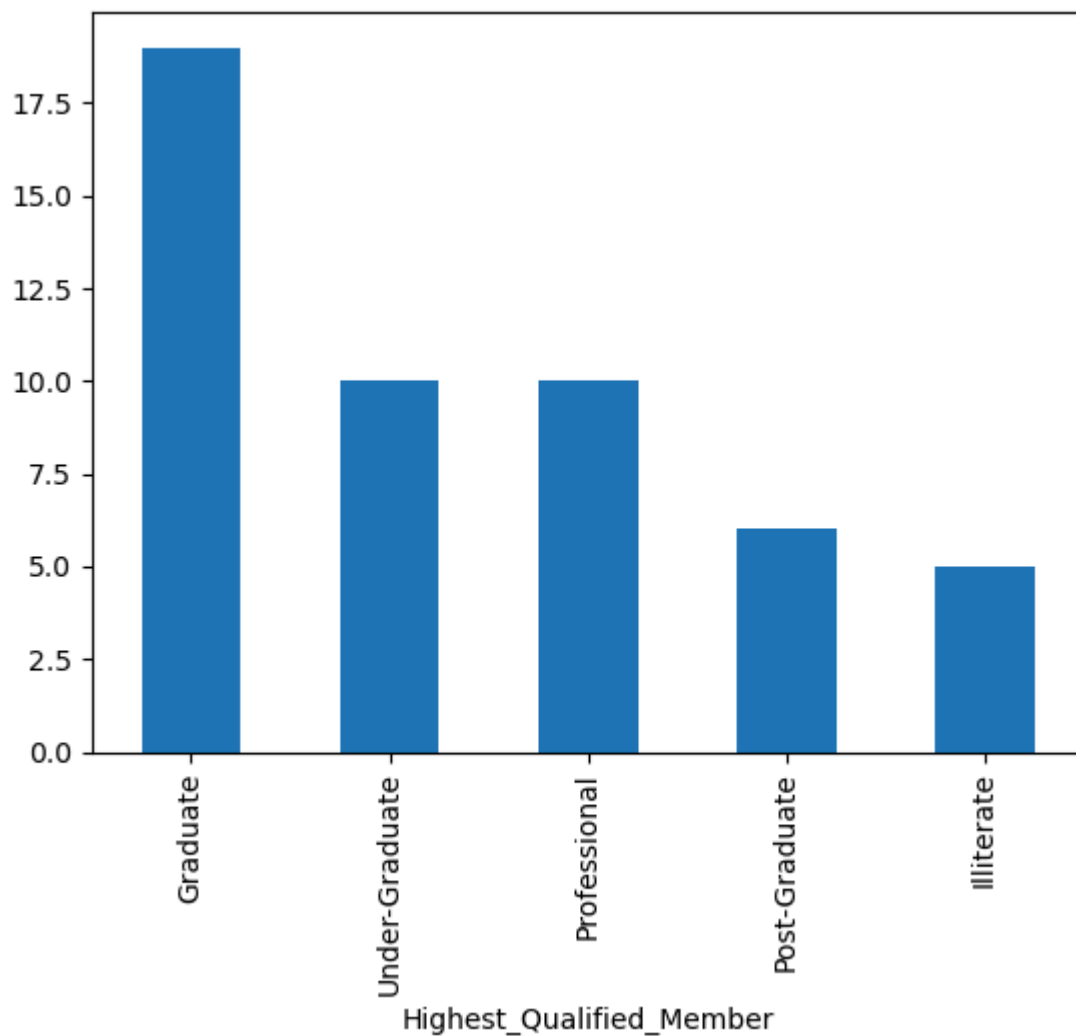
col_0	Mthly_HH_Expense	count
18	25000	8

```
In [28]: income_df.columns
```

```
Out[28]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',  
              'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',  
              'No_of_Earning_Members'],  
              dtype='object')
```

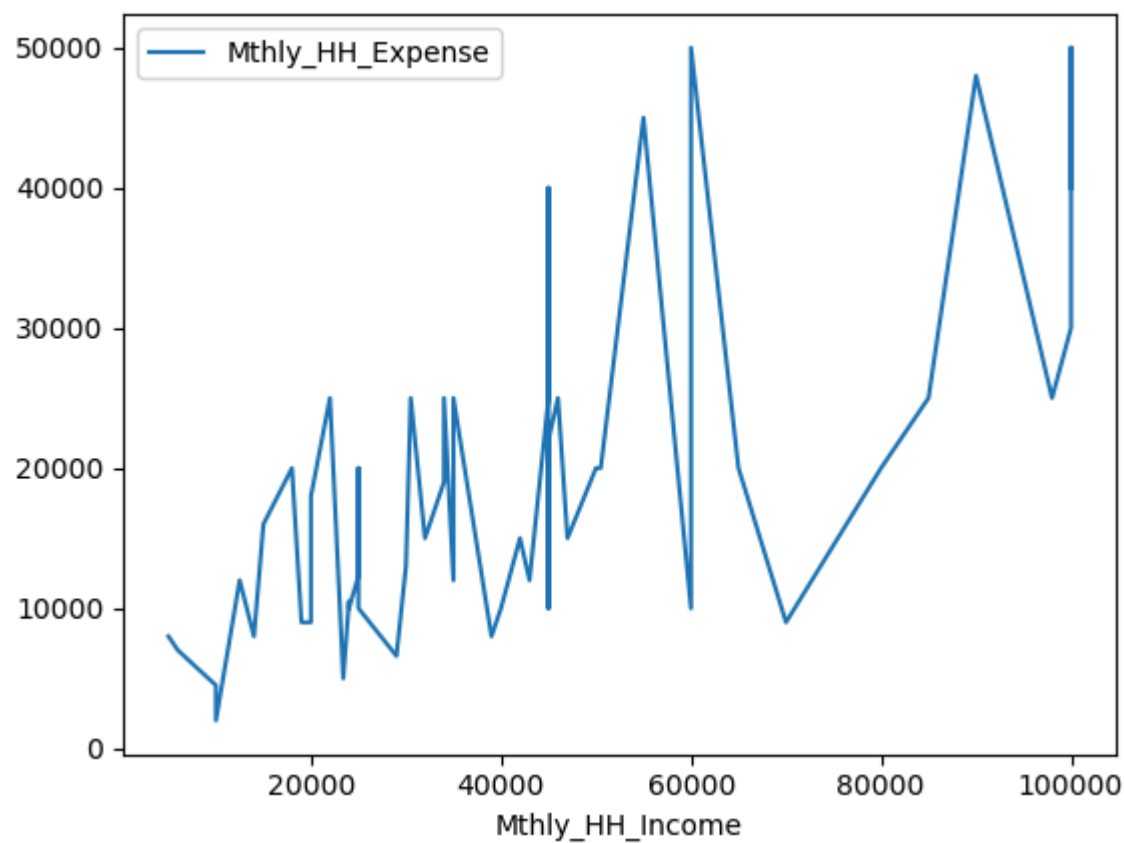
```
In [30]: income_df['Highest_Qualified_Member'].value_counts().plot(kind = 'bar')
```

```
Out[30]: <Axes: xlabel='Highest_Qualified_Member'>
```



```
In [33]: income_df.plot(x = 'Mthly_HH_Income', y = 'Mthly_HH_Expense')
```

```
Out[33]: <Axes: xlabel='Mthly_HH_Income'>
```

```
In [35]: print(income_df['Mthly_HH_Expense'].quantile(0.75))
```

25000.0

```
In [36]: print(income_df['Mthly_HH_Expense'].quantile(0.50))
```

15500.0

```
In [37]: print(income_df['Mthly_HH_Expense'].quantile(0.25))
```

10000.0

```
In [44]: IQR = income_df['Mthly_HH_Expense'].quantile(0.70) - income_df['Mthly_HH_Expense'].quantile(0.30)
IQR
```

Out[44]: np.float64(12899.999999999993)

```
In [42]: IQR = income_df['Mthly_HH_Expense'].quantile(0.50) - income_df['Mthly_HH_Expense'].quantile(0.25)
IQR
```

Out[42]: np.float64(5500.0)

```
In [45]: income_df['Mthly_HH_Expense'].var()
```

Out[45]: 146173342.85714287

```
In [46]: income_df['Mthly_HH_Expense'].std()
```

Out[46]: 12090.216824240286

```
In [48]: coef = income_df['Mthly_HH_Expense'].std() / income_df['Mthly_HH_Expense'].mean()
print(coef)
```

0.6424814977277227

```
In [50]: income_df['Annual_HH_Income'].skew
```

```
Out[50]: <bound method Series.skew of 0 64200
```

```
1      79920
2     112800
3      97200
4     147000
5     196560
6     167400
7     216000
8     218880
9     220800
10    278400
11    279840
12    292032
13    316800
14    244800
15    246000
16    261000
17    258000
18    348000
19    385200
20    351360
21    445440
22    330480
23    469200
24    466200
25    449400
26    556920
27    412800
28    488880
29    619200
30    523800
31    507600
32    437400
33    610200
34    596160
35    456840
36    570000
37    581760
38    600600
39    590400
40    590400
41    647400
42    756000
43   1075200
44   1142400
45    885600
46   1152480
47   1404000
48   1032000
49   1320000
```

```
Name: Annual_HH_Income, dtype: int64>
```

```
In [52]: print(income_df['Annual_HH_Income'].kurtosis())
```

```
1.1012908548129197
```

```
In [ ]:
```